Econ 272 / GSB 607 **Department of Economics**
Spring 2025 **Stanford University**

Problem Set V

Due: Sunday May 25^{th}, 2025, 6pm.

1. Here we will be using the Jacob-Lefgren data. The running variable is an the minimum of the prior reading score rdge0 and the prior math score mtge0. Students are eligiable for summer school if the running variable is less than 2.75. The outcome of interest is the later math score, rmath1. The treatment is participation in summer school, summer.

   (a) What fraction participates in the summer program if eligible, and what fraction participates if not eligible?

   (b) Estimate the average effect of participation in the summer program using the minimum of the two scores as the running variable using local linear regression. How do you select the bandwith?

   (c) Suppose you only use the mtge0 score as the running variable. How does that change your estimates, and how does that change your interpretation?

   (d) Suppose you only use the rdge0 score as the running variable. How does that change your estimates, and how does that change your interpretation?

   (e) Use one of the methods relying on weight estimation, using the minimum of the two scores as the running variable. How do the results change?